MMC Watershed Rainfall Report

Rainfall observations for the Moores Mill Creek watershed

MMC Watershed Data

August 7, 2026

MMC Watershed Data collects and standardizes rainfall observations from Auburn and Opelika stations near the Moores Mill Creek watershed. This report presents station observations and a Thiessen area-weighted watershed rainfall series for one saved collection selected at render time.

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1 Purpose and reporting period

MMC Watershed Data retrieves rainfall observations from the City of Auburn LI-COR network and the City of Opelika THOR network, preserves the API responses as evidence, validates their structure, and converts them into comparable station CSVs. This report uses the **latest saved MMC collection**, with the reporting period read from its saved processed CSV filenames. The selected period is **May 01, 2026 through August 07, 2026**, with **114,948 validated observations** from **9 stations**. All configured stations are represented. It asks: **what total and temporal rainfall pattern does Thiessen weighting estimate for the MMC watershed during this period?**

2 Station and watershed location

The Moores Mill Creek (MMC) watershed is the land area where surface runoff drains toward Moores Mill Creek. Its boundary defines the geographic area of interest for this project. Figure 1 places the configured Auburn and Opelika rainfall stations relative to that boundary: blue points are Auburn stations, red points are Opelika stations, and the green polygon is the watershed.

The colored areas inside the green boundary are the clipped Thiessen influence regions used for watershed weighting. Each station measures rainfall at one location. Stations inside and around the watershed help show how rainfall varies across the area, but an individual station value should not be interpreted as rainfall over the entire watershed.

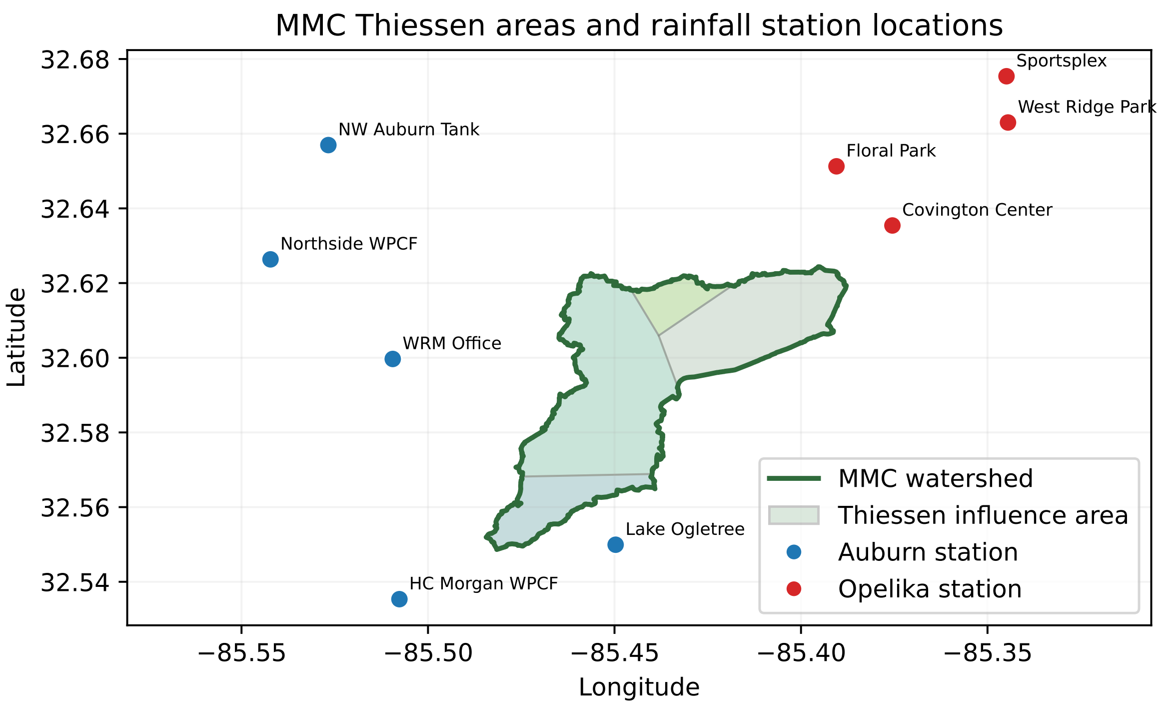

Figure 1: Thiessen influence areas and rainfall stations relative to the Moores Mill Creek watershed boundary.

3 Data and method

The report selects one processed collection using the date range encoded in MMC's station filenames. It always chooses the collection most recently written under `data/processed/`, so its period follows the latest API collection made with `mmc`. If no generated collection exists, the report uses the bundled dataset under `reports/data/` so the repository PDF remains reproducible offline.

All records are loaded through `mmc_watershed_data.reporting`, which calls the project's typed `load_rainfall_records` function. The report does not recreate Auburn epoch conversion, Opelika cumulative-counter conversion, Pydantic validation, or API requests. Rainfall totals are sums of processed interval `RainIn` values in inches. The processed CSVs already contain one `RainIn` value per source observation at a nominal 10-minute cadence. The station time-series figures plot those rows directly at their saved `Date_hour` timestamps; the report does not resample or sum them into new time windows.

For watershed-scale rainfall, the report calls the project's shared spatial module. It projects WGS 84 station and watershed geometries to WGS 84 / UTM Zone 16N (`EPSG:32616`), clips nearest-station Thiessen regions to the MMC watershed, and uses each clipped area fraction as a fixed station weight. Stations need at least 90% of the expected nominal intervals for the period. Timestamps are aligned to the nearest 10-minute label only for this derived calculation. Distinct Auburn or Opelika timestamps assigned to one label are summed to preserve interval rainfall. Exact repeated rows are counted once; divergent values at an identical timestamp remain a station-level quality conflict. Original processed rows remain unchanged. If any positive station weight is missing at a label, the watershed value remains blank and is marked incomplete; weights are not redistributed and missing rainfall is not replaced with zero.

4 Results

4.1 Summary

For **May 01, 2026 through August 07, 2026**, **Floral Park** recorded the largest station total at **25.53 inches**. Across complete intervals, the Thiessen watershed rainfall totals **15.08 inches**. Auburn stations contributed **40.71 inches** and Opelika stations contributed **97.60 inches** when station interval values are summed.

4.2 Station statistics

Table 1: Processed rainfall statistics by station for the selected reporting period.

Station	City	Total rain (in)	Wet rows	Max interval (in)
Floral Park	Opelika	25.53	492	0.81
Sportsplex	Opelika	24.29	477	0.87
West Ridge Park	Opelika	24.06	469	0.60
Covington Center	Opelika	23.72	485	0.69
NW Auburn Tank	Auburn	9.99	505	0.31
WRM Office	Auburn	9.95	477	0.28
HC Morgan WPCF	Auburn	9.46	457	0.25
Northside WPCF	Auburn	9.13	466	0.34
Lake Ogletree	Auburn	2.19	108	0.23

4.3 Thiessen watershed rainfall

The projected MMC watershed area is **30.04 km2**. **8 stations** met the 90% period threshold and **1** were excluded. The series contains **13,852 complete** and **404 incomplete** 10-minute intervals.

Table 2: Period-level station eligibility, clipped watershed area, and fixed Thiessen weights.

Station	Eligible	Coverage	Area (km2)	Watershed (%)	Weight	Aggregated	Exclusion
WRM Office	True	99.5%	14.589	48.57	0.48569	3	
Lake Ogletree	False	14.4%	0.000	0.00	0.00000	2	temporal coverage below 90%
Northside WPCF	True	99.5%	0.000	0.00	0.00000	2	
NW Auburn Tank	True	99.6%	0.000	0.00	0.00000	0	
HC Morgan WPCF	True	99.6%	4.113	13.69	0.13694	0	
Sportsplex	True	98.7%	0.000	0.00	0.00000	1	
Floral Park	True	98.8%	2.035	6.78	0.06775	0	

Station	Eligible	Coverage	Area (km2)	Watershed (%)	Weight	Aggregated	Exclusion
West Ridge Park	True	98.8%	0.000	0.00	0.00000	1	
Covington Center	True	97.3%	9.300	30.96	0.30962	1	

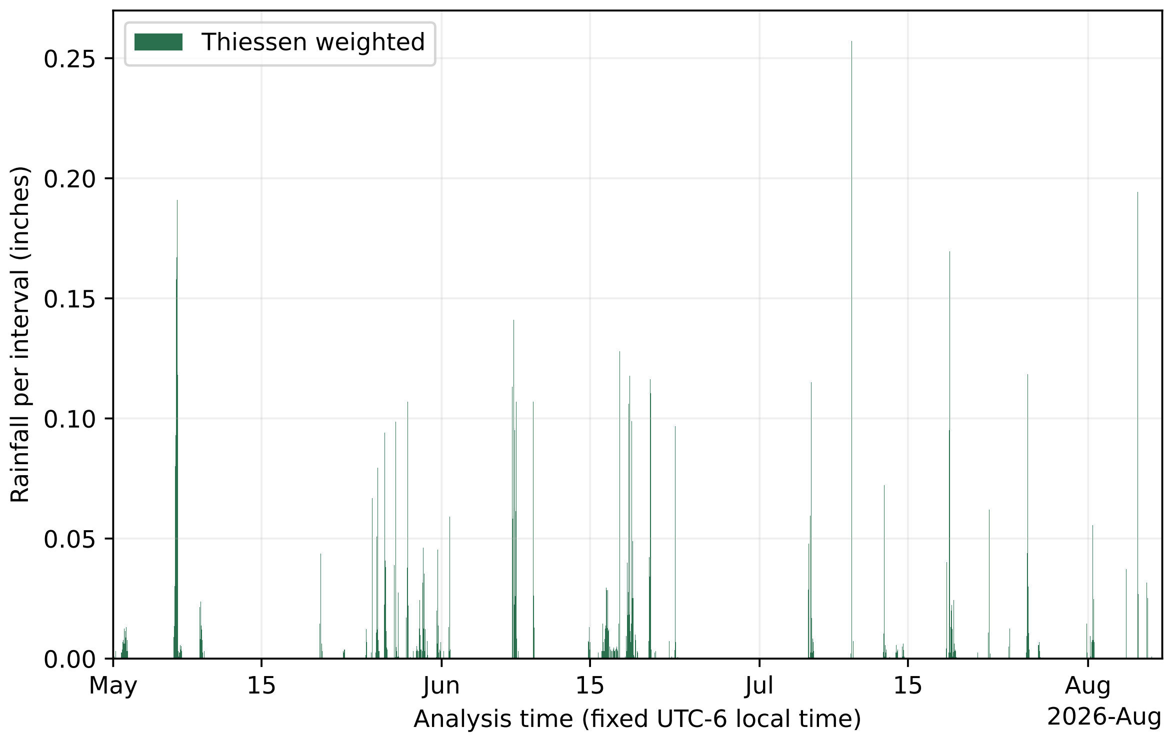

Figure 2: Thiessen-weighted MMC watershed rainfall at complete 10-minute intervals.

4.4 Rainfall totals

Blue bars identify Auburn stations and red bars identify Opelika stations, matching the city colors used by the dashboard. The legend makes that encoding explicit.

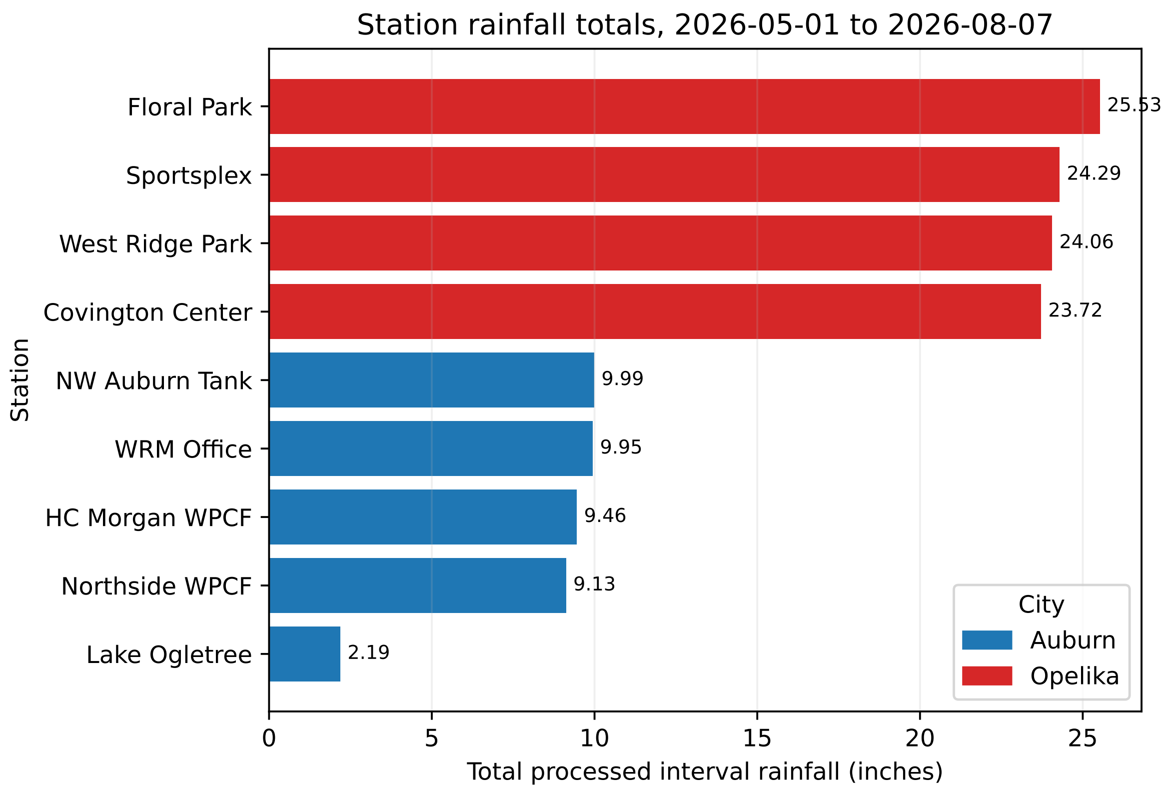

Figure 3: Total processed interval rainfall by station for the selected reporting period.

4.5 Rainfall at 10-minute intervals

Each panel below represents one station over the full selected period. Every bar reports one `RainIn` value directly from the processed CSV; the shared horizontal scale makes the timing of rainfall comparable across stations.

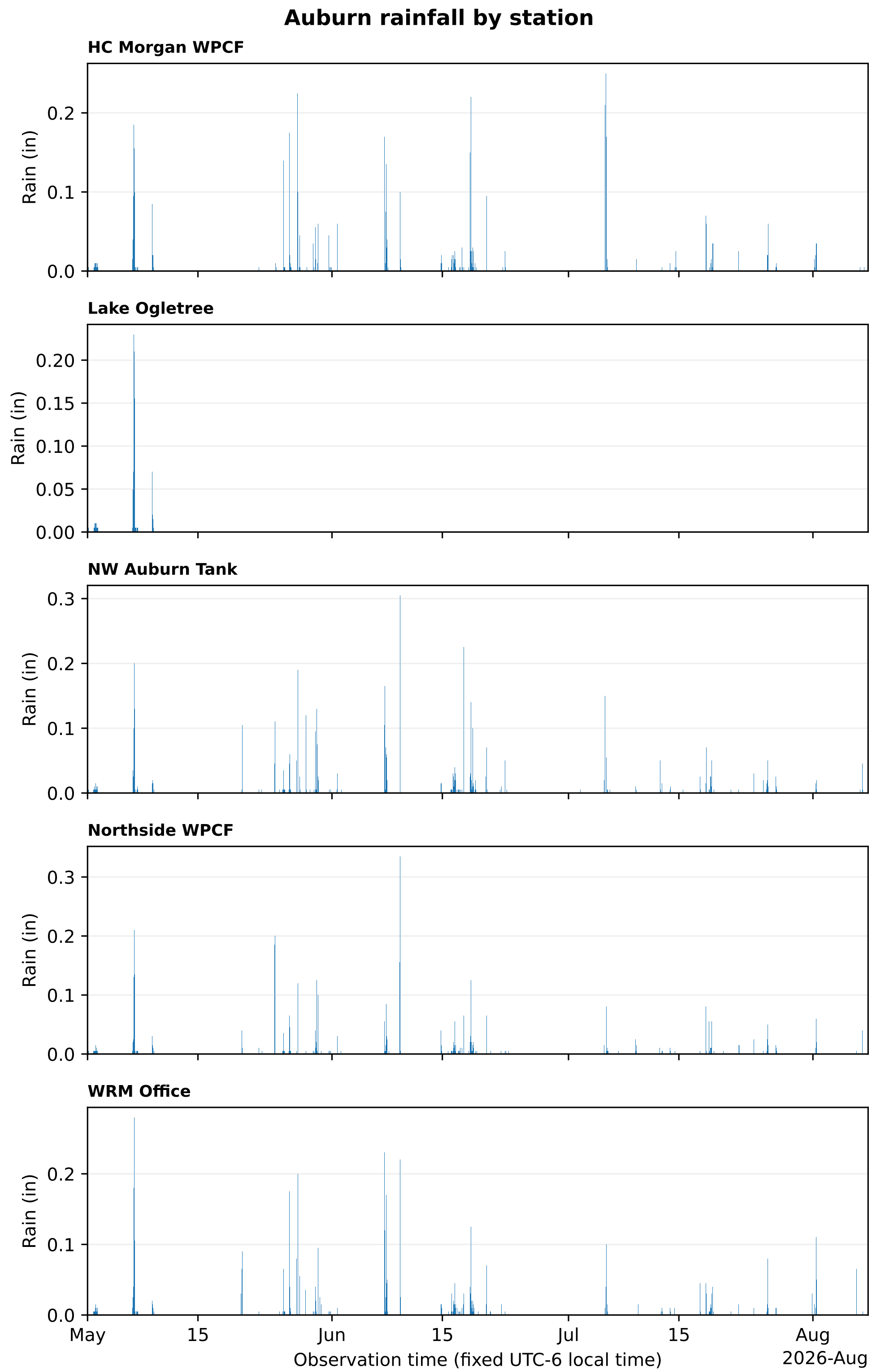

Figure 4: Auburn processed rainfall observations at a nominal 10-minute cadence during the selected reporting period.

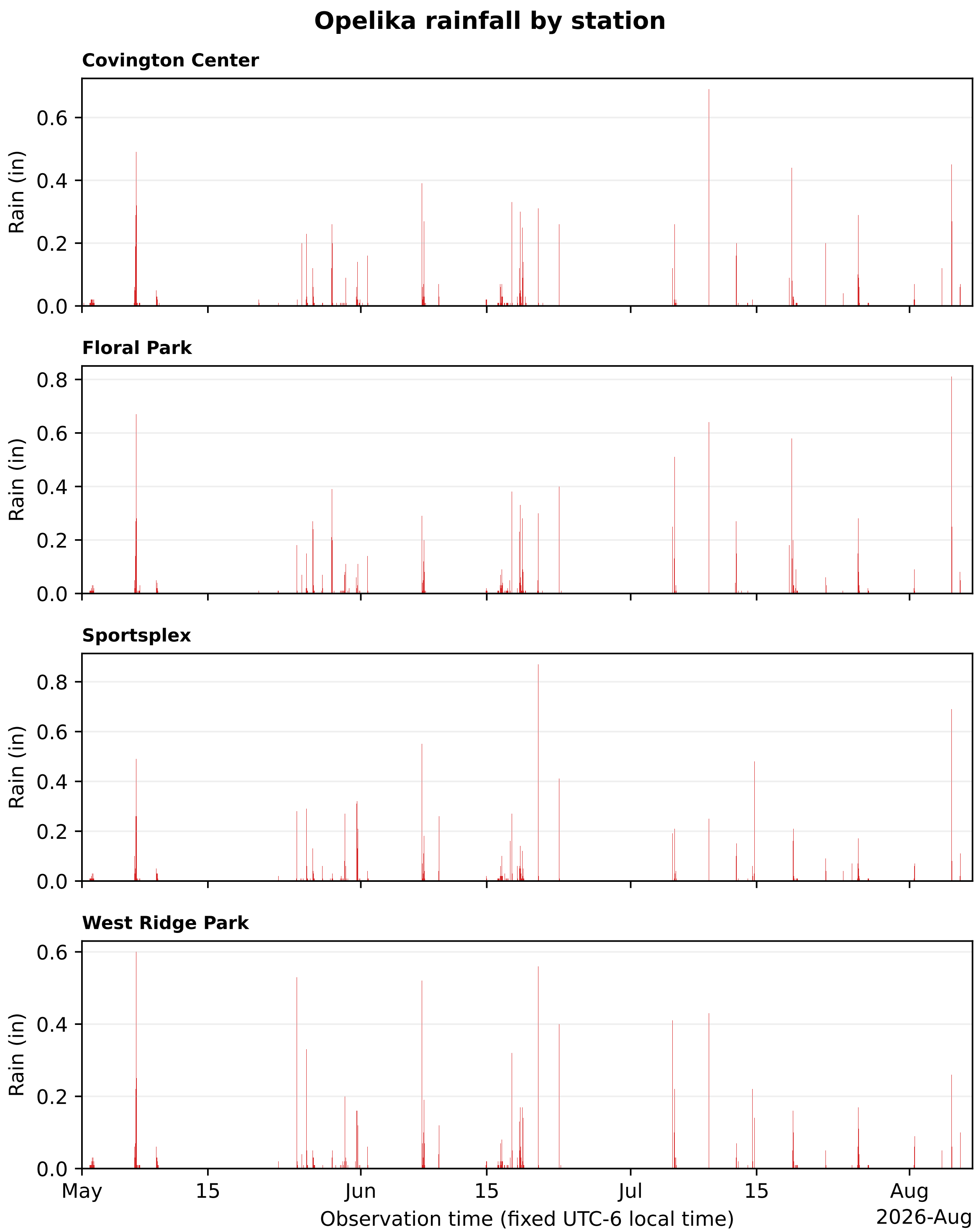

Figure 5: Opelika processed rainfall observations at a nominal 10-minute cadence during the selected reporting period.

4.6 Interpretation and limitations

The selected records show that **Floral Park** had the largest accumulated interval total during **May 01, 2026 through August 07, 2026**. The The Thiessen series converts eligible point observations into a watershed estimate by nearest-station area. The complete intervals represent **15.08 inches** of watershed rainfall. Only 13,852 complete intervals should be used as model input without an explicit future gap-filling decision. Thiessen rainfall is a nearest-station spatial estimate, not radar rainfall or proof that precipitation was uniform inside each polygon.

5 Provenance and reproducibility

Table 3: Data and software provenance recorded for this report render.

Field	Recorded value
Selected input	latest saved MMC collection
Requested period	2026-05-01 through 2026-08-07
Processed station files	9
Validated observations	114948
Stations represented	9
Spatial method	Thiessen nearest-station area weighting
Spatial analysis CRS	EPSG:32616
Station coverage threshold	90% of nominal 10-minute intervals
Rainfall unit	inches per processed interval
Auburn source	https://www.licor.cloud/api/v2/timeseriesdata
Opelika source	https://360.thormobile.net/thorcloud/api/weatherpacketsbyinterval
Project version	0.6.0
Python	3.13.11
Rendered (UTC)	2026-08-08T00:54:43+00:00

After collecting a period, the report automatically selects the most recently written processed collection:

```
uv run mmc --start-date 2026-01-01 --end-date 2026-01-08
quarto render reports/mmc-rainfall-report.qmd --to pdf
```

The report source, inputs, PDF, metadata, and logs contain no API key or other secret.